

Curriculum Vitae

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Primož Potočnik,

Faculty of Mathematics and Physics,
University of Ljubljana, Jadranska 19,
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Education, Scientific Degrees and Academic Titles:

- 2000, Ph. D., University of Ljubljana.
(Thesis: Graph Symmetries and Group Actions;
advisor: Prof. Dragan Marušič.)
- 1998, M. Sc. in Mathematics, University of Ljubljana.
(Thesis: Some Classical Results on Permutation Groups;
advisor: Prof. Dragan Marušič.)
- 1995, B. Sc. in Mathematics, University of Ljubljana.
(Thesis: Vertex-transitive Graphs Which Are Not Cayley;
advisor: Prof. Dragan Marušič.)

Work Experience:

- October 2012 – present: Full Professor,
Department of Mathematics, Faculty of Mathematics and Physics, University
of Ljubljana, Slovenia.
- October 2007 – 2012: Associate Professor,
Department of Mathematics, Faculty of Mathematics and Physics, University
of Ljubljana, Slovenia.
- October 2005 – September 2007: Lecturer,
Department of Mathematics, Faculty of Mathematics and Physics, University
of Ljubljana, Slovenia.
- September 2004 – September 2005: Postdoctoral Fellow,
The University of Auckland – Department of Mathematic, Auckland, New
Zealand.
- August 2003 – January 2004: Postdoctoral Fellow,
University of Ottawa – Department of Mathematics and Statistics, Ottawa,
Canada.
- September 2002 – present: Adjoint Research Fellow,
Institute of Mathematics, Physics and Mechanics, University of Ljubljana,
Slovenia.
- October 2001 – August 2002: Assistant Professor,
Faculty of Education – Department of Mathematics and Computer Science,
University of Ljubljana, Slovenia.

- December 1995 – September 2001: Sessional Lecturer, Department of Mathematics, Faculty of Mathematics and Physics, University of Ljubljana, Slovenia.

Research Interests:

Algebraic combinatorics, permutation groups, graph theory.

Administration:

- Head of Department of Mathematics, Faculty of Mathematics and Physics, University of Ljubljana, 2017–2019, 2025–present.
- Vice-Dean For Teaching, Faculty of Mathematics and Physics, University of Ljubljana, 2017–2019, 2025–present.
- Member of the *Council of the European Mathematical Society*, 2018–2022, 2024–2028.
- President of the *Society of Mathematicians, Physicists and Astronomers of Slovenia*, 2022–2024.
- Deputy Head of Department of Mathematics, Faculty of Mathematics and Physics, University of Ljubljana, 2007–2009.

Member of editorial board:

- Discrete Mathematics (member of the editorial board, since 2020),
- Ars mathematica contemporanea (member of the editorial board), 2012–2023 <https://amc-journal.eu/index.php/amc/issue/archive>, <http://amc.imfm.si/index.php/amc>
- The art of discrete and applied mathematics (member of the editorial board) since 2018–2023, <https://adam-journal.eu/index.php/ADAM/index>
- Discrete applied mathematics (visiting editor for a special issue), 2017, <http://www.journals.elsevier.com/discrete-applied-mathematics/>.

Member of programme or organising committee at mathematical meetings:

- Chair of the organising and member of the scientific committee, *10th Slovenian Conference on Graph Theory*, Kranjska Gora, 18th–24th June 2023.
- Cofounder of “Algebraic Graph Theory International Webinar (AGTIW)”, every two weeks from November 2020 onwards, link: <http://euler.doa.fmph.uniba.sk/AGTIW.html>.
- Member of the organising committee, 8th European Mathematica Congress, Portorož, July 2021 (until June 2020).
- Member of the programme committee, *EUROCOMB 2019*, Bratislava, August 2019.
- Member of the organising committee, *8th Slovenian Conference on Graph Theory*, Kranjska Gora, 21st–27th June 2015.
- Member of the scientific committee, *Algorithmic Graph Theory on the Adriatic Coast*, Koper, 16th–19th June 2015.

- Member of the organising committee, *Ljubljana-Leoben Seminar 2012*, Bovec, 20th–22th September 2012.
- Member of programme committee, *Workshop on Graph Asymmetries*, Massey University, Palmerston North, New Zealand, 23. – 25. February 2005.

Longer visits of universities and research institutions:

- *University of Auckland*, Auckland, New Zealand, January – March 2023.
- *Northern Arizona University*, Flagstaff, USA, February 2004 – July 2004;
- *Univeristy of Ottawa*, Ottawa, Canada, August 2003 – January 2004;
- *University of Western Australia*, Perth, Australia, April – June 2003;
- *University of Auckland*, Auckland, New Zealand, February – April 2003.

Awards, honors and grants:

- *ARRS research programme*, 2020–2027, 2,8 FTE per year.
- *ARRS research grant*, 2021–2024, 0,7 FTE (50.000 EUR) per year.
- *ARRS research grant*, 2019–2022, 1,4 FTE (100.000 EUR) per year.
- *ARRS research grant*, 2011–2014, 2 FTE per year.
- *ARRS research grant*, 2008–2010, 1 FTE per year.
- *Fulbright Scholarship*, 5 months, academic year 2003/04.

Publications:

Scientific articles, Published or Accepted for Publication

1. M. Barbieri, M. Lekše, P. Potočnik, K. Rekvényi, Separating subsets from their images, *Forum of Mathematics, Sigma*, **14** (2026), e111.
<https://doi.org/10.1017/fms.2026.10258>
2. G. Cunningham, A. Montero, E. Mochán, P. Potočnik, M. Toledo, Faithfulness in maniplexes, *J. Alg. Combin.*, to appear.
3. R. J. Evans, P. Potočnik, A census of Cayley graphs, *J. Symb. Comp.* **137** (2026), 102580.
<https://doi.org/10.1016/j.jsc.2026.102580>
4. N. Bašić, P. W. Fowler, B. T. Pickup, P. Potočnik, Random walks and the electronic structure of graphene, *J. Chem. Physics* **164** (2026), 104103.
<https://doi.org/10.1063/5.0319282>
5. N. Bašić, P. W. Fowler, M. M. McCarthy, P. Potočnik, Nut digraphs, *Discrete Applied Math.* **383** (2026), 203–226.
<https://doi.org/10.1016/j.dam.2025.12.037>
6. I. Hubard, P. Potočnik, P. Šparl, An infinite family of simple graphs underlying chiral, orientable reflexible and non-orientable rotary maps, *J. Combin. Theory Ser. B* **176** (2026), 304–403.
<https://doi.org/10.1016/j.jctb.2025.10.001>

7. M. Conder, P. Potočnik, Edge-transitive cubic graphs: analysis, cataloguing and enumeration, *J. Algebra* **685** (2026), 703–737.
<https://doi.org/10.1016/j.jalgebra.2025.07.035>
8. P. Potočnik, M. Toledo, A. Ramos, S. Wilson, Transitive cornerations in maps, *Electronic J. Combin.*, **32(4)**, (2025), #P4.4.
<https://doi.org/10.37236/12798>.
9. A. Montero, P. Potočnik, Vertex-transitive graphs with small motion and transitive permutation groups with small minimal degree, *J. Combin. Theory Ser. A* **216**, article no. 106065 (2025),
<https://doi.org/10.1016/j.jcta.2025.106065>.
10. M. Conder, P. Potočnik, A note on normal covers of 2-arc-transitive graphs, *Algebraic Combinatorics* **8** (2025), 469–478
<https://doi.org/10.5802/alco.415>
11. M. Barbieri, P. Potočnik, Vertex-primitive digraphs with large fixity, *Annali di Matematica*, **203** (2024), 2383–2403.
<https://doi.org/10.1007/s10231-024-01447-x>
12. P. Potočnik, M. Toledo, G. Verret, *On orders of automorphisms of vertex-transitive graphs*, *J. Combin. Theory Ser. B.* **166** (2024), 123–153.
13. M. Conder, L. Morgan, P. Potočnik, Resolution of a conjecture about linking ring structures, *Journal of Algebra*, **632** (2023), 87–101.
14. P. Potočnik, M. Toledo, Cubic vertex-transitive graphs admitting automorphisms of large order, *Bulletin of Malaysian Mathematical Sciences Society*, **46** (2023) art. 133 (33 pages)
15. P. Potočnik, J. Vidali, Cubic vertex-transitive graphs of girth 6, *Discrete Mathematics* **345** (2022), 112734;
16. R. Jajcay, P. Potočnik, S. Wilson, On the Cayleyness of Praeger-Xu graphs, *J. Combin. Theory Ser. B.* **152** (2022), 55–79.
17. A. Hujdurović, P. Potočnik, G. Verret, Three local actions in 6-valent arc-transitive graphs, *J. Graph Theory* **99** (2022), 207–216.
18. F. Lehner, P. Potočnik, P. Spiga, On fixity of arc-transitive graphs, *Science China Mathematics* **64** (2021), 2603–2610.
19. P. Potočnik, P. Spiga, *On the number of fixed points of automorphisms of vertex-transitive graphs*, *Combinatorica* **41** (2021), 703–747.
20. P. Potočnik, P. Spiga, *On minimal degree of transitive permutation groups with stabiliser being a 2-group*, *J. Group Theory* **24** (2021), 619–624.
21. P. Potočnik, M. Toledo, *Generalised voltage graphs*, *European J. Combin.* **94** (2021), 103313, 20pp.
22. P. Potočnik, M. Toledo, *Finite cubic graphs admitting a cyclic group of automorphism with at most three orbits on vertices*, *Discrete Mathematics* **344** (2021), 112195.
23. P. Potočnik, G. Verret, S. Wilson, *Base Graph – Connection Graph: Dissection and Construction*, *Discrete Applied Math.* **291** (2021), 116–128.

24. R. Jajcay, P. Potočnik, S. Wilson, Half-Cyclic, Dihedral and Half-Dihedral Codes, *J. Applied Math. and Comp.* **64** (2020), 691–708.
25. P. Potočnik, M. Toledo, *Classification of cubic vertex-transitive tricirculants*, *Ars Math. Contemp.* **18** (2020), 1–31.
26. P. Potočnik, S. Wilson, *Recipes for Edge-Transitive Tetravalent Graphs*, *The Art of Discrete and Applied Mathematics* **3** (2020), #P1.08.
27. P. Potočnik, J. Vidali, *Girth-regular graphs*, *Ars Math. Contemp.* **17** (2019), 249–368.
28. R. Jajcay, P. Potočnik, S. Wilson, *The The Praeger-Xu Graphs: Cycle Structures, Maps and Semitransitive Orientations*, *Acta Mathematica Universitatis Comenianae* **88** (2019), 269–291.
29. P. Potočnik, P. Spiga, *Lifting a prescribed group of automorphisms of graphs*, *Proc. Amer. Math. Soc.* **147** (2019), 3787–3796.
30. D. Pellicer, P. Potočnik, M. Toledo, *An existence result on two-orbit maniplaxes*, *J. Combin. Theory Ser. A.* **166** (2019), 226–253.
31. L. Morgan, P. Potočnik, G. Verret, *Arc-transitive digraphs of given out-valency and with blocks of given size*, *J. Combin. Theory Ser. B.* **137** (2019), 118–125.
32. R. Möller, P. Potočnik, N. Seifter, *Infinite arc-transitive and highly-arc-transitive digraphs* *European J. Combin.*, **77** (2019), 78–89.
33. R. Möller, P. Potočnik, N. Seifter, *Sharply k -arc-transitive-digraphs: finite and infinite examples*, *Discrete Math.*, **342**, (2019), 768–776.
34. P. Potočnik, S. Wilson, *Linking rings structures and semisymmetric graphs: combinatorial constructions*, *Ars Math. Contemp.* **14** (2018), 1–17.
35. B. Kuzman, A. Malnič, P. Potočnik, *Tetravalent Vertex- and Edge-Transitive Graphs Over Doubled Cycles*, *J. Combin. Theory Ser. B.* **131** (2018), 109–137.
36. P. Potočnik, P. Šparl, *On the radius and the attachment number of tetravalent half-arc-transitive graphs*, *Discrete Math.* **340** (2017), 2967–2971.
37. P. Potočnik, S. Wilson, *The separated box product of two digraphs*, *European J. Combin.* **62** (2017), 35–49.
38. P. Potočnik, P. Spiga, G. Verret, *Asymptotic enumeration of vertex-transitive graphs of fixed valence*, *J. Combin. Theory Ser. B* **122** (2017), 221–240.
39. P. Potočnik, R. Požar, *Smallest tetravalent half-arc-transitive graphs with the vertex-stabiliser isomorphic to the dihedral group of order 8*, *J. Combin. Theory, Ser. A.* **145** (2017), 172–183.
40. K. Berčič, P. Potočnik, *Two-arc-transitive two-valent digraphs of certain orders*, *Ars Math. Contemp.* **11** (2016), 127–146.
41. P. Potočnik, S. Wilson, *Linking rings structures and semisymmetric graphs: Cayley constructions*, *European J. Combin.* **51** (2016), 84–98.
42. P. Potočnik, P. Spiga, G. Verret, *Bounding the order of the vertex-stabiliser in 3-valent vertex-transitive and 4-valent arc-transitive graphs*, *J. Combin. Theory, Ser. B.* **111** (2015), 148–180.

43. P. Potočnik, P. Spiga, G. Verret, *A census of 4-valent half-arc-transitive graphs and arc-transitive digraphs of valence two*, Ars Math. Contemp. **8** (2015), 133–148.
44. M. Conder, P. Potočnik, P. Šparl, *Some recent discoveries about half-arc-transitive graphs*, Ars Math. Contemp. **8** (2015), 149–162.
45. A. Malnič, P. Potočnik, N. Seifter, P. Šparl, *Reachability relations, transitive digraphs and groups*, Ars Math. Contemp. **8** (2015), 83–94.
46. M. Giudici, L. Morgan, P. Potočnik, G. Verret, *Elusive groups of automorphisms of digraphs of small valency*, European J. Combin. **46** (2015), 1–9.
47. M. Conder, C.-H. Li, P. Potočnik, *On the orders of arc-transitive graphs*, J. Algebra **421** (2015), 167–186.
48. M. Giudici, P. Potočnik, G. Verret, *Semiregular automorphisms of edge-transitive graphs*, J. Alg. Combin. **40** (2014), 961–972.
49. M. Knor, M. Mačaj, P. Potočnik, R. Škrekovski, *Trees T satisfying $W(L^3(T)) = W(T)$* , Filomat **28** (2014), 551–556.
50. P. Potočnik, P. Spiga, G. Verret, *On the order of arc-stabilisers in arc-transitive graphs with prescribed local group*, Trans. Amer. Math. Soc. **366** (2014), 3729–3745.
51. M. Knor, M. Mačaj, P. Potočnik, R. Škrekovski, *Complete solution of equation $W(L^3(T)) = W(T)$ for Wiener index of iterated line graphs of trees* Disc. Appl. Math. **171** (2014), 90–103.
52. M. Knor, P. Potočnik, R. Škrekovski, *Relationship between edge-Wiener index and Gutman index of a graph*, Disc. Appl. Math. **167** (2014), 197–201.
53. P. Potočnik, *Tetravalent arc-transitive locally-Klein graphs with long consistent cycles*, European J. Combin. **36** (2014), 270–281.
54. P. Potočnik, S. Wilson *Linking ring structures and tetravalent semisymmetric graphs*, Ars Math. Contemp. **7** (2014), 341–352.
55. P. Potočnik, *Locally arc-transitive graphs of valence $\{3, 4\}$ with trivial edge kernel* J. Alg. Combin. **38** (2013), 637–651.
56. V. Andova, M. Knor, P. Potočnik, R. Škrekovski, *On a variation of Randić index*, Australasian J. Combin. **56** (2013), 61–75.
57. M. Knor, P. Potočnik, *A note on 2-subset-regular self-complementary 3-uniform hypergraphs*, Ars Combinatoria **111** (2013), 33–36.
58. M. Knor, P. Potočnik, R. Škrekovski, *Wiener index of iterated line graphs of trees homeomorphic to H* , Discrete Math. **313** (2013), 1104–1111.
59. M. Knor, P. Potočnik, R. Škrekovski, *Wiener index of iterated line graphs of trees homeomorphic to the claw $K_{1,3}$* , Ars Mathematica Contemporanea **6** (2013), 211–219.
60. P. Potočnik, P. Spiga, G. Verret, *Cubic vertex-transitive graphs on up to 1280 vertices*, J. Symbolic Comp. **50** (2013), 465–477.
61. M. E. Lladser, P. Potočnik, J. Širáň, M. C. Wilson, *The diameter of random Cayley digraphs of given degree*, Disc. Math. and Theoretical Comp. Science **14** (2012), 83–90.

62. M. Knor, P. Potočnik, R. Škrekovski, *Wiener index in iterated line graphs*, Discrete Applied Math., **160** (2012), 2234–2245.
63. M. Dutour Sikirić, M. Knor, P. Potočnik, R. Škrekovski, J. Širáň, *Hyperbolic analogues of fullerenes on orientable surfaces*, Discrete Math. **312** (2012), 729–736.
64. M. Knor, P. Potočnik, R. Škrekovski, *On a conjecture about Wiener index in iterated line graphs of trees*, Disc. Math **312** (2012), 1094–1105.
65. M. Knor, P. Potočnik, *Efficient domination in cubic vertex-transitive graphs*, European J. Combin **33** (2012), 1755–1764.
66. P. Potočnik, P. Spiga, G. Verret, *On the nullspace of arc-transitive graphs over finite fields*, J. Alg. Combin. **36** (2012), 389–401.
67. P. Potočnik, P. Spiga, G. Verret, *On graph-restrictive permutation groups*, J. Combin. Theory, Ser. B., **102** (2012), 820–831.
68. P. Potočnik, S. Wilson, *On the point-stabiliser in a transitive permutation group*, Monatshefte für Mathematik **166** (2012), 497–504.
69. P. Potočnik, P. Spiga, G. Verret, *Tetravalent arc-transitive graphs with unbounded vertex-stabilisers*, Bull. Australian Math. Soc., **84** (2011), 79–89.
70. M. Boben, Š. Miklavič, P. Potočnik, *Rotary polygons in configurations*, Electron. J. Combin. **18** (2011), paper no. 119, 1–19.
71. P. Potočnik, G. Verret, *On the vertex-stabiliser in arc-transitive digraphs*, J. Combin. Theory Ser. B. **100** (2010), 497–509.
72. M. Conder, P. Potočnik, J. Širáň, *Regular maps with almost Sylow-cyclic automorphism groups, and classification of regular maps with Euler characteristic $-p^2$* , J. Algebra **324** (2010), 2620–2635.
73. A. Orbanić, M. Petkovšek, T. Pisanski in P. Potočnik, *A note on enumeration of one-vertex maps*, Ars mathematica contemporanea **3** (2010), 1–12.
74. P. Potočnik and M. Šajna, *Brick assignments and homogeneous almost self-complementary graphs*, J. Combin. Theory, Ser. B. **99** (2009), 185–201.
75. P. Potočnik, *A list of 4-valent 2-arc-transitive graphs and finite faithful amalgams of index $(4, 2)$* European. J. Combin. **30** (2009), 1323–1336.
76. P. Potočnik and M. Šajna, *Vertex-transitive self-complementary uniform hypergraphs*, European J. Combin. **30** (2009), 327–337.
77. M. Boben, Š. Miklavič, P. Potočnik, *Consistent cycles in $\frac{1}{2}$ -arc-transitive graphs*, Electronic J. Combin. **16** (2009), # R5, 1–10.
78. P. Potočnik, M. Šajna, *The existence of regular self-complementary 3-uniform hypergraphs*, Discrete Mathematics **309** (2009), 950–954.
79. M. Conder, P. Potočnik, J. Širáň, *Regular hypermaps over projective linear groups*, J. Austral. Math. Soc. **85** (2008), 155–175.
80. Š. Miklavič, P. Potočnik, S. Wilson, *Arc-transitive cycle decompositions of tetravalent graphs*, J. Combin. Theory, Ser. B **98** (2008), 1181–1192.
81. M. Giudici, C. H. Li, P. Potočnik, and C. Praeger, *Homogeneous factorisations of graph products*, Disc. Math. **308** (2008), 3652–3667.

82. P. Potočnik and M. Šajna, *A note on transitive permutation groups of degree twice a prime*, Ars Mathematica Contemporanea **1** (2008), 165–168.
83. Š. Miklavič, P. Potočnik, S. Wilson, *Overlap in consistent cycles*, J. Graph Theory **55** (2007), 55–71.
84. P. Potočnik, S. Wilson, *Tetravalent edge-transitive graphs of girth at most 4*, Journal of Combinatorial Theory Ser. B., **97** (2007), 217–236.
85. Š. Miklavič, P. Potočnik, *Distance-regular Cayley Graphs on Dihedral Groups*, Journal of Combinatorial Theory Ser. B., **97** (2007), 14–33.
86. Š. Miklavič, P. Potočnik, and S. Wilson, *Consistent cycles in graphs and digraphs*, Graphs and Combinatorics **23** (2007), 205–216.
87. P. Potočnik, M. Šajna, *Self-complementary two-graphs and almost self-complementary double covers*, European J. Combin. **28** (2007), 1561–1574.
88. A. Malnič, Š. Miklavič, D. Marušič, P. Potočnik, *Semisymmetric elementary abelian covers of the Möbius-Kantor graph*, Discrete Mathematics **307** (2007), 2156–2175.
89. P. Potočnik, M. Šajna, G. Verret, *Mobility of vertex-transitive graphs*, Discrete Mathematics **307** (2007), 579–591.
90. M. Giudici, C. H. Li, P. Potočnik, C. Praeger, *Homogeneous factorisations of complete multipartite graphs*, Discrete Mathematics **307** (2007), 415–431.
91. Š. Miklavič, P. Potočnik, *Distance-transitive Dihedrants*, Designs, and Cryptography **41** (2006), 185–193.
92. A. Malnič and P. Potočnik, *Invariant subspaces, duality, and covers of the Petersen graph*, European J. Combin. **27** (2006), 971–989.
93. M. C. E. Conder, A. Malnič, D. Marušič, P. Potočnik, *A census of cubic semisymmetric graphs on up to 768 vertices*, J. Alg. Combin. **23** (2006), 255–294.
94. P. Potočnik and M. Šajna, *On almost self-complementary graphs*, Disc. Math. **306** (2006), 107–123.
95. M. Giudici, C. H. Li, P. Potočnik, and C. Praeger, *Homogeneous factorisations of graphs and digraphs*, European J. Combin. **27** (2006), 11–37.
96. P. Potočnik, M. Škoviera, R. Škrekovski, *Nowhere-zero 3-flows in Cayley graphs of abelian groups*, Discrete Mathematics **297** (2005), 119–127.
97. M. C. E. Conder, A. Malnič, D. Marušič, T. Pisanski, P. Potočnik, *The cubic edge- but not vertex-transitive graph on 112 vertices*, J. Graph Theory **50** (2005), 25–42.
98. P. Potočnik, *Edge-colourings of cubic graphs admitting a solvable vertex-transitive group of automorphisms*, Journal of Combinatorial Theory Ser. B, **91** (2004), 289–300.
99. A. Malnič, D. Marušič, P. Potočnik, *On cubic graphs admitting an edge-transitive solvable group*, Journal of Algebraic Combinatorics **20** (2004), 99–113.
100. A. Malnič, D. Marušič, P. Potočnik, *Elementary abelian covers of graphs*, Journal of Algebraic Combinatorics **20** (2004), 71–97.

101. A. Malnič, D. Marušič, P. Potočnik, C. Q. Wang, *An infinite family of cubic edge- but not vertex-transitive graphs*, Discrete Mathematics **280** (2004), 133–148.
102. Š. Miklavic and P. Potočnik, *Distance-regular circulants*, European J. Combin. **24** (2003), 777–784.
103. D. Marušič, P. Potočnik, *Bridging semisymmetric and half-arc-transitive actions on graphs*, European J. Combin. **23** (2002), 719–732.
104. P. Potočnik, *On 2-arc-transitive Cayley graphs of abelian groups*, Discrete Mathematics **244** (2002), 417–421.
105. D. Marušič, P. Potočnik, *Classifying 2-arc-transitive graphs of order a product of two primes*, Discrete Mathematics **244** (2002), 331–338.
106. P. Potočnik, *B-groups of order a product of two primes*, Mathematica Slovaca **51** (2001), 63–67.
107. D. Marušič, P. Potočnik, *Semisymmetry of generalized Folkman graphs*, European J. Combin. **22** (2001), 333–349.

Submitted for Publication

108. M. Conder, G. Potočnik, P. Potočnik, Computing the Hamiltonian compression factors of cubic graphs,
<https://arxiv.org/abs/2606.21941>
109. P. Potočnik, M. Toledo, A. Ramos, S. Wilson, Orbits of consistent walks in dart-transitive maps,
<https://arxiv.org/pdf/2503.14241>
110. M. Conder, P. Potočnik, Nilpotence of automorphism groups of discrete objects
111. R. Jajcay, M. Lekše, P. Potočnik, Š. Glevitzká, Cubic girth-regular graphs of girth 6

Book chapters

112. P. Potočnik, P. Spiga, G. Verret, *Groups of order at most 6 000 generated by two elements, one of which is an involution, and related structures* in “Symmetries in Graphs, Maps, and Polytopes; 5th SIGMAP Workshop, West Malvern, UK, July 2014” (eds. J. Širáň, R. Jajcay), Springer Proceedings in Mathematics and Statistics (2016).
113. T. Pisanski, P. Potočnik, Chapter *Graphs on surfaces* in “Handbook of graph theory”, The CRC Press series on discrete mathematics and its applications (2004).

Other articles:

114. P. Potočnik, *Symmetry – in Graph Theory and in General* (Slovenian), Obzornik za matematiko in fiziko, **49**, (2002), 87–95.
115. P. Potočnik, *Counting - Basic and Less Basic* (Slovenian), Obzornik za matematiko in fiziko, **48**, (2001), 65–80.

116. P. Potočnik, *Electromagnetic potential of a charged moving particle* (Slovenian), *Obzornik za matematiko in fiziko*, **42**, (1995), 72–79.

Invited Lectures and Plenary Talks:

- “Restricted Symmetry in Graphs and Maps” (plenary talk), Symmetries in Graphs, Maps and Polytopes (SIGMAP) 2026, Brussels, 13th–17th July 2026.
- “Datasets of regular maps and their skeletons” (plenary talk), Graph Embeddings and Maps on Surfaces 2025, Trenčianske Teplice, Slovakia, 22nd–27th June 2025, <https://gems2025.savbb.sk>
- “Generating and Cataloging Symmetric Graphs” (plenary talk), Information Technologies – Applications and Theory 2024, Drienica, Slovakia, 20th – 24th September 2024, <https://ceur-ws.org/Vol-3792/>.
- “Fixity of Graphs” (invited talk), Symmetries of Discrete Objects, Rotorua, New Zealand, 10th –14th February 2020, <https://www.math.auckland.ac.nz/~conder/SOD0-2020/>
- “Some problems about symmetries of finite graphs” (invited talk), Graphs, groups, and more: celebrating Brian Alspach’s 80th and Dragan Marušič’s 65th birthdays, 27th May – 1st June 2018, <https://conferences.famnit.upr.si/indico/event/4/material/slides/21.pdf>
- International Workshop on Symmetries of Graph and Networks (invited talk), 29th January – 2nd February 2018, Sanya, China, <http://www.tsimf.cn/meeting/show?id=83#MMM4>
- “Cubic vertex-transitive graphs” (invited talk), Leoben-Ljubljana GraphTheory Seminar 2016, September 2016, Judenburg, Austria.
- “Censuses of graphs and maps of certain symmetry types” (invited talk), 5th Workshop SIGMAP – Symmetries In Graph, Maps And Polytopes, 7th – 11th July 2014, West Malvern, U.K.
- “How many symmetries can a symmetric graph have?” (plenary talk), 37th Australian Conference on Combinatorial Mathematics and Combinatorial Computing: the University of Western Australia, December 9–13, 2013. Perth.
- “Edge-colourings of Cubic Vertex-transitive Graphs” (invited talk), *Grafy 2009*, Terchova, Slovakia, (May 2009).
- “Graphs and their symmetries” *Colloquium of the Department of Mathematics and Statistics, Northern Arizona University*, Flagstaff, 5th February 2008.
- “Graf, grupe in amalgami”, *Matematični kolokvij DMFA*, Ljubljana, 27th September 2005.
- “Symmetry in graphs”, *Combinatorics and Optimization Seminar*, University of Ottawa, Canada, 15 September 2003.
- “Regular Covers of Graphs and Their Symmetries”, *Algebra, Geometry and Combinatorics Seminar Series*, The University of Auckland, New Zealand, 10 March 2003.
- “Elementary Abelian Covers of Graphs”, *Groups and Combinatorics Seminar*, The University of Western Australia, Perth, Australia, 12 May 2003.

- “Flows and colourings of vertex-transitive graphs”, *Bratislavsky Seminar z Teorie Grafov*, Comenius University, Bratislava, Slovakia, September 2001;